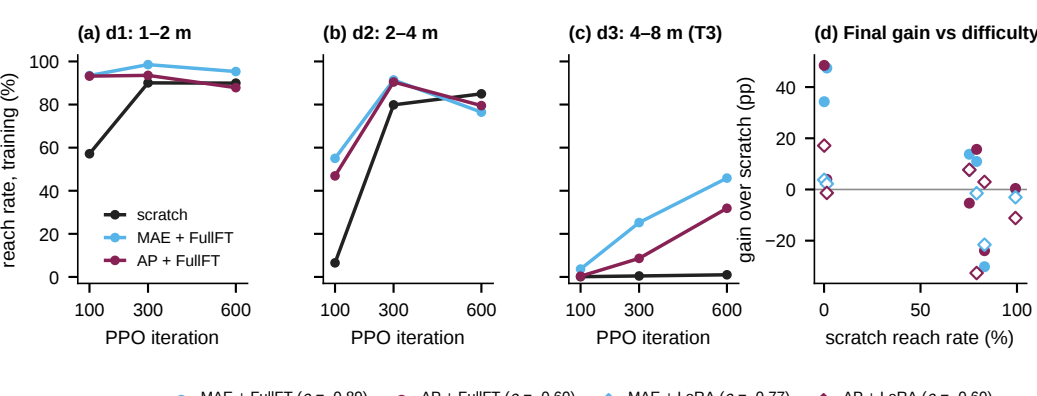
(a) d1: 1–2 m
(b) d2: 2–4 m
(c) d3: 4–8 m (T3)
(d) Final gain vs difficulty
reach rate, training (%)
gain over scratch (pp)
PPO iteration
PPO iteration
PPO iteration
scratch reach rate (%)
scratch
MAE + FullFT
AP + FullFT
MAE + FullFT (ρ = -0.89)
AP + FullFT (ρ = -0.60)
MAE + LoRA (ρ = -0.77)
AP + LoRA (ρ = -0.60)